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COMPARISON OF REAL-TIME PRESSURE RAIL SELECTION ALGORITHMS FOR THE HYBRID HYDRAULIC ELECTRIC ARCHITECTURE: CASE STUDY ON A TRACK LOADER

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ABSTRACT

The hybrid hydraulic electric architecture (HHEA) seeks to combine the high power/torque/force density of hydraulics with the efficiency of electric machines. A set of common pressure rails is used to provide a majority of the power and this power is modulated by small electric machines to provide precise control for the operator. The HHEA has been studied in previous work using off-line dynamic programming optimization to determine energy efficient pressure rail selections, but this approach requires drive cycle information apriori. A Lagrange multiplier method has also been investigated where a set of gains (Lagrange multipliers) are optimized off-line with the idea the these gains, once determined, could be used for real-time operation. In this work, three new real-time pressure rail selection algorithms that do not require future drive cycle information are investigated; greedy, torque minimizing, and thresholding. The greedy control is found to only use 1% more energy than the globally optimal dynamic programming solution; but a model of energy loss is required.

Keywords: HHEA, Mobile Machines, Digital Hydraulics, Common Pressure Rails

NOMENCLATURE

β	Fluid bulk modulus
Δt	Time duration of each time step
$\eta_{P/M}$	Efficiency of the main pump in pumping or motoring
$\frac{dV}{dt}$	Rate of volume expansion due to actuator movement
λ	Lagrange multiplier
A_{cap}	Hydraulic acting area of the cap side of either work cylinder
A_{rod}	Hydraulic acting area of the rod side of either work cylinder
$d(\cdot)$	Trajectory of pressure rail and motor displacement selections
D_m	Full displacement of the propel hydraulic motor
D_{pm}	Displacement of the hydraulic pump/motor for modulating work force
F_{elect}	Electric force produced to modulate the work circuit forces
J	Total cost function
J_{DP}	Dynamic programming cost function
J_G	Greedy algorithm cost function
J_{LM}	Lagrange multiplier cost function
J_{TM}	Torque minimizing control cost function
K	Size of switching valve oriface ($Q = k\sqrt{(\Delta P)}$)
k_f	Number of time steps considered
$l^{P/M}$	Loss per flow in the main pump if motoring or pumping
$Loss_{elec,motor}$	Energy loss in the propel electric motor
$Loss_{hyd,motor}$	Energy loss in the propel hydraulic motor
$Loss_{MP,i}$	Energy loss in the main pump due to the ith rail
$Loss_{switch}$	Energy loss due to throttling in the valves
P	The pressure on the actuator side of the control valves
$P_{A/B}$	Inlet/outlet pressure of the propel hydraulic motor
P_{cap}	Pressure of the cap side of either work cylinder
P_{high}	Pressure of the highest pressure rail
P_{low}	Pressure of the lowest pressure rail
P_{mid}	Pressure of the middle pressure rail
P_{rod}	Pressure of the cap side of either work cylinder
P_{tank}	Pressure of the hydraulic reservoir
PR	The set of pressures of each rail
Q_i	Flow out of rail i
T_{elect}	Torque transmitted by the propel electric motor
T_{pm}	Torque provided by the electric motor/generator for modulating work force
V	Volume on the actuator side of the control valves
V_i	Net flow out of rail i
x	Fractional displacement of the propel hydraulic motor

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1. INTRODUCTION

Off road vehicles (construction, agriculture, and mining machines) typically utilize diesel engine driven hydraulic circuits because of the high torque and power requirements of these applications. These machines are estimated to account for about 1% of total U.S. energy use[1]. The environmental effects of these machines can be reduced by making these machines more efficient and electric powered.

Current fully electric loaders utilize electric motors to drive the propel functions and either electric motor driven lead screws or electro-hydraulic actuators (EHAs) to drive the lift and tilt functions[2–4]. These approaches force each electric motors to be able to handle both the max force and velocity (corner power) of their actuator, making the electric motors very large and the machines prohibitively expensive compared to diesel machines.

The hybrid hydraulic-electric architecture (HHEA) is a solution to the problem of electrifying these vehicles at a lower cost (by requiring smaller electric motors and batteries). It was originally proposed in [5] to combine the advantages of hydraulics (torque and power density) with the benefits of electric machines (controllability and efficiency). A set of common pressure rails (CPRs) are used to transmit most of the power hydraulically and small electric motors are used to add or remove small amounts of power for precise control. The HHEA can save 50-80% of energy compared to a load sensing circuit, depending on the vehicle and task [6]. This work will focus specifically on a 5 ton compact track loader for which we are developing fully electric powered version of HHEA. For this machine, the corner powers of the electric motors used for power modulation in the HHEA can be downsized by almost 11 times compared to the direct electrification strategy for the propel function. For the lift and tilt functions, the HHEA can downsize the corner power by 4-6 times.

A key aspect of operating this architecture is the selection of hydraulic pressure rails at either end of each actuator. The selection of these rails determines the amount of power required by the electric modulation motor/generators in order to meet the trajectory commanded by the driver. It is not obvious how to select between these pressure rails at any point in time. Two methods have previously been studied for the HHEA. The first was the Lagrange multiplier method [7] which requires the full set of drive cycle data statistics to optimize the Lagrange multipliers. Once the Lagrange multipliers are obtained, it can be used to select pressure rails in real time by optimizing an instantaneous cost-function. However, switching losses must be ignored in this framework. Hence, a second, fully off-line optimization framework was constructed using dynamic programming in order to incorporate switching losses and find the global optimal solution [8]. As dynamic programming requires complete drive cycle data in order to select efficient pressure rail combinations, it cannot be used in real-time. In this paper, we introduce three other possible directly real-time implementable pressure rail selection algorithms. These control algorithms are compared against the global optimal dynamic programming solution in simulation using typical drive cycle data from a 5 ton compact track loader to determine their appropriateness to be used in real-time while minimizing energy loss.

The HHEA is not the only system architecture that utilizes multiple common pressure rails. The STEAM architecture, introduced in 2013 [9], was one of the earlier uses of CPRs. More recently, a multiple variable CPRs architecture was also proposed in [10] for planters. Both of these architectures still use throttling to buck pressure to the required level, but the CPR pressures are varied with time in the later architecture. CPR's were selected using a greedy control law where the instantaneous losses at each time are minimized. There have been other CPR architectures proposed outside of mobile machines, like the power take-off of the WaveStar wave energy converter which utilized CPRs and multi-area actuators [11]. There, a thresholding control is utilized where the CPRs are only changed once the difference between the desired force and the discrete hydraulic force exceeds some threshold [12]. The three new controllers investigated here for the HHEA are a) a greedy control, b) a torque minimizing control, and c) a thresholding control.

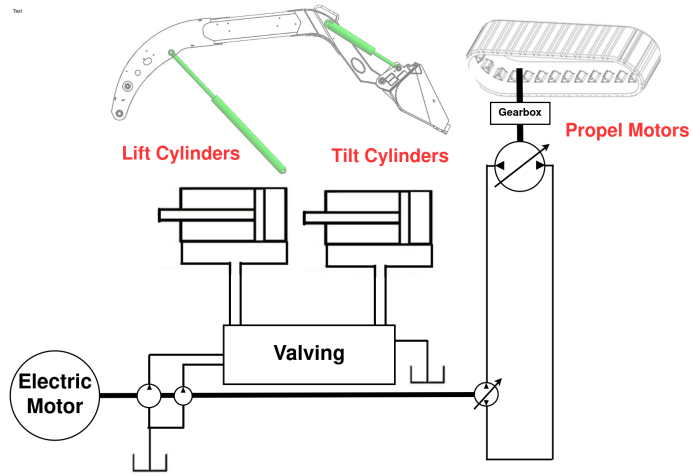

FIGURE 1: BASELINE ARCHITECTURE FOR USED ON THE STOCK COMPACT TRACK LOADER MACHINE

This rest of the paper is organized as follows: a background into the application and the HHEA is presented in section 2, the system model for the HHEA is presented in section 3, each controller is presented in section 4, results are presented in section 5, and concluding remarks are made in section 6.

2. BACKGROUND

The architecture of the stock compact track loader is shown in figure 1. It features two main hydraulic circuits: the work circuit (which powers the movement of the arm and bucket) and the propel circuit (which powers the tracks that move the vehicle chassis). The work circuit is powered by two gear pumps. The flow from these pumps is throttled down to meet the drive cycle demand. The work circuit has the largest opportunity for efficiency improvement. This circuit accounts for less than 15% of the output work but more than half of the losses of the baseline machine. The propel circuit of the stock machine uses a pair of hydrostatic circuits to power the two tracks. A two displacement axial-piston motor powers each track through a 20:1 gear box. The motors are fed by variable displacement pumps driven

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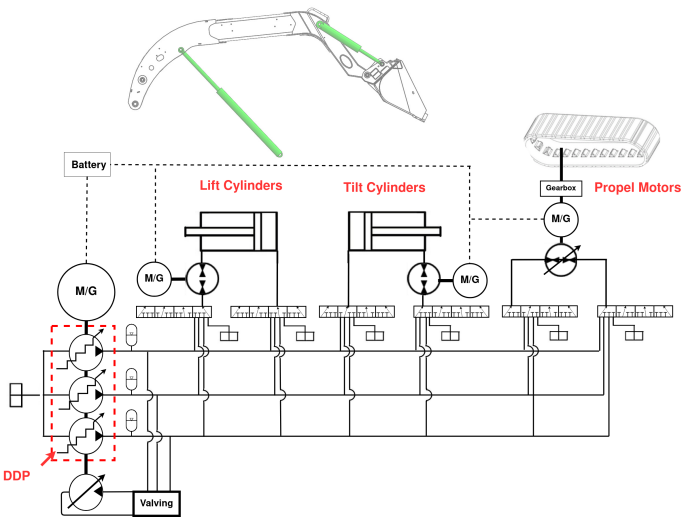

FIGURE 2: SCHEMATIC OF THE HYBRID HYDRAULIC-ELECTRIC ARCHITECTURE (HHEA) FOR A COMPACT TRACK LOADER. ONLY ONE OF PROPEL FUNCTIONS IS SHOWN FOR SIMPLICITY.

at constant speed. In the stock machine, a joystick commands the displacement of the pumps and a button controls the motor displacements.

The proposed HHEA circuit for the compact track loader is shown in Fig. 2. There are 4 pressure rails at substantially constant pressures, common for both the work circuit (tilt and lift linear actuators), and the propel circuit (left and right hydraulic motors). The pressure rails could be maintained by a main pump (which also may operate as a motor at times). The main pump could be constructed in a variety of ways, but the architecture shown here includes a digital displacement pump (DDP) coupled with a hydraulic motor. As the current version of the DDP can only pump so a separate motor is included to remove flow from the rails is the pressure in any rail gets too high. Besides being very efficient, the DDP has 3 outlet ports, or "pumplets", making it function essentially as three separate variable displacement pumps on the same shaft.

The same or similar propel hydraulic motors and gear boxes and work circuit hydraulic actuators in the stock machine are used in the redesigned system. In the work circuit, a small electric motor/generator drives a small hydraulic pump/motor to buck or boost the pressure from the rails in order to meet the drive cycle demand. For the propel circuit, a small electric motor adds or subtracts torque from the shaft of the hydraulic motor. Switching valves are used to select which CPRs to connect to either end of each actuator (or to the inlet of the hydraulic pump motor in the case of the rod end of the work cylinders). It is the control of these valves that is the focus of this paper. The pressurized CPRs are provided by the 3 pumplets of a digital displacement pump (DDP). A hydraulic motor is used to occasionally absorb the return flows from the CPRs, since, as mentioned eariler, the current DDP cannot be motored.

Force/torque is generated via a combination of hydraulic and electric force/torque. The selected CPRs determine the hydraulic force/torque and the remainder required to achieve the drive cycle is made up by the electric motor/generators. For the work circuit, the drive cycle force is achieved by

$$F = \underbrace{(P_{cap}A_{cap} - P_{rod}A_{rod})}_{\text{hydraulic force}} + F_{elect} \tag{1}$$

where the port pressures P_{cap} and P_{rod} are chosen from the set of pressure rails $PR = \{P_{high}, P_{mid}, P_{low}, P_{tank}\}$. A_{cap} and A_{rod} are the hydraulic areas of the cap and rod sides of the actuator. F_{elect} is produced by applying an electric torque on the hydraulic pump/motor: $F_{elect} = \frac{2\pi A_{rod}}{D_{pm}} T_{mg}$ where D_{pm} is the displacement of the small hydraulic pump/motor and T_{mg} is the electric torque provided by the small electric motor/generator. For the propel circuit, the drive cycle torque is achieved by

$$T = \underbrace{x\frac{D_m}{2\pi}(P_A - P_B)}_{\text{hydraulic torque}} + T_{elect} \tag{2}$$

where the port pressures P_A, P_B are chosen from PR, the CPR pressures, $x = \{0.65, 1\}$ is the binary displacement fraction option of the motor and D_m is the maximum displacement. For both work and propel circuits, it is assumed here that T_{elect} perfectly makes up the difference between the hydraulic force/torque and the drive cycle requirement. The pressure rail selection algorithm involves selecting P_{cap} and P_{rod} (for both lift and tilt), P_A, P_B, and x (for both the left and right tracks). These decisions are made every 200 ms.

3. SYSTEM MODEL

Losses are considered from each hydraulic pump/motor and electric/generator as well as from throttling across the control valves.

The energy losses due to the hydraulic pump/motors and electric motor/generators are found using loss maps which are static lookup tables based on operating conditions. These look-up tables were given by the manufacturers. For real-time deployment, these look-up tables would be converted into polynomial expressions for the sake of computational efficiency.

The main pump's efficiency and hence loss per unit of flow are dependent on the pump speed and pressure, and whether it is pumping or motoring. The power loss per unit flow for the ith pressure rail are:

$$\begin{aligned} l_i^P &= PR_i\left(\frac{1}{\eta_P(\omega, PR_i)} - 1\right) \\ l_i^M &= PR_i\left(\eta_M(\omega, PR_i) - 1\right) \end{aligned} \tag{3}$$

where $\eta_P(\omega, PR_i)$ and $\eta_M(\omega, PR_i)$ are the pumping and motoring efficiencies at the rail pressure i. No losses are incurred by taking flow or returning flow to the tank pressure rail. The energy loss in the main pump depends on the sign of each rail's net flow V_i,

$$Loss_{MP,i}(k, d(\cdot)) = \Delta t \begin{cases} l_i^P Q_i(k, d(k)) & \text{if } V_i \geq 0 \\ l_i^M Q_i(k, d(k)) & \text{if } V_i < 0 \end{cases} \tag{4}$$

where $d(k)$ is a trajectory of pressure rail and motor displacement selections, Δt is the amount of time in one decision time step (200

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ms), and the flow in rail i at time step k is denoted by $Q_i(k, d(k))$. The volume in each rail at the end of the simulation is given as $V_i = \sum_0^{k_f} Q_i(k, d(k))\Delta t$.

The losses due to switching are calculated based on the pressure dynamics on the actuator side of the valve. Switches occur separately at each side of each actuator. There are two sides to each actuator, and there are 4 actuators (1 lift cylinder, 1 tilt cylinder, and two propel motors). Switching losses depend on the present and next pressure rail selections as well as the actuator position and flow demands. For details see [8].

All losses are calculated only when decisions are made (every Δt seconds), except for switching losses, which are computed on a much smaller time resolution in order to better approximate continuous time. The pressures of the each side (rod/cap or inlet/outlet) of actuator (lift, tilt, left track, right track) are given by

$$\dot{P}(t) = -\frac{\beta}{V(t)}\left(\frac{dV}{dt}(t) - \sum_{i=1}^{4} Q_i(t)\right) \tag{5}$$

where β is the bulk modulus of the working fluid, $\frac{dV}{dt}(t)$ is the rate of volume expansion due to the actuator movement, and Q_i is the flow from rail i to the actuator (or the inlet of the hydraulic pump/motor in the case of the rod side of the lift and tilt cylinders). The volume in the outlet of the valve changes over time only in the case of valves connected to the cap side of the lift and tilt actuators. The flow from each rail is given by the orifice equation:

$$Q_i(t) = Kx_i(t)sgn(\Delta P_i(t)))\sqrt{|\Delta P_i(t)|} \tag{6}$$

where $\Delta P_i = PR_i - P$, and K is a sizing parameter for the switching valves, and x_i represents the fraction of orifice that is open for the ith inlet port at time t. We assume that the valve spool trajectory $x_i(t)$ follows a second order system response with a damping ratio of 0.7 followed by saturation at zero and full valve opening. A delay is included between the closing of one valve and the opening of another. This delay is determined via a grid search of delays and selecting the delay which results in the lowest throttling loss without cavitating the fluid. This grid search is performed at many combinations of conditions such that a look-up table is generated: for a given actuator volume, rate of volume change, initial pressure rail, and final pressure rail, the near optimal delay can be selected.

The power loss at one side of an actuator for switching between pressure rails is found by the energy loss through each valve:

$$Loss_{switch,j}(k, d(k), , d(k+1)) = \sum_{i=1}^{4}\int_0^{\Delta t} Q_i(t)\Delta P_i(t)dt \tag{7}$$

and the total energy loss is found by considering each side of each actuator (i.e. index j corresponds to the cap and rod sides of the lift and tilt actuators as well as the inlet and outlet ports of the left and right propel actuators).

$$Loss_{switch}(k, d(k), d(k+1)) = \sum_{j=\{cap/rod/in/out\}} Loss_{switch,j}(k, d(k), d(k+1)) \tag{8}$$

The entire expression of loss can now be written as:

$$J(d(\cdot)) = \sum_{k=0}^{k_f}\Bigg[Loss_{hyd,motor}(k, d(k)) + Loss_{e-motor}(k, d(k)) + \sum_{i=2}^{4} Loss_{MP,i}(k, d(\cdot)) + Loss_{switch}(k, d(k), , d(k+1))\Bigg] \tag{9}$$

where $d(k)$ is a trajectory of pressure rail and motor displacement selections, k_f is the number of time steps considered, $Loss_{hyd,motor}$ is the loss in the hydraulic motors, $Loss_{e-motor}$ is the loss in the electric motors, $Loss_{MP,i}$ is the loss in the main pump due to the ith rail, and $Loss_{switch}$ is the throttling loss. Note that $Loss_{MP,i}$ depends on the assumed net flow direction on each rail in Eq. (4).

4. RAIL SELECTION CONTROL ALGORITHMS

The 5 control algorithms for selecting pressure rails studied in this work are: *Lagrange multiplier method*, *dynamic programming algorithm*, *greedy control law*, *torque minimizing control law*, and *thresholding*. The decision criteria cost functions used in the execution of each control law may differ, but each algorithm is compared to each other using the total loss cost function in Eq. (9). The real-time controllers need only be able to be computed every 50 – 200 milliseconds. The real-time controllers presented here should be simple enough to run on these time scales, but this assumption has not tested.

4.1 Dynamic Programming

Dynamic programming (DP) allows for a global solution to the cost function in (9) that must be found off-line by considering full drive-cycle knowledge ahead of time.

Because the main pump losses cannot be formulated unless the net rail flow directions are known, care must be taken for this term. For each rail, the possible net flow directions are positive (requiring pumping), negative (requiring motoring) or zero (requiring no energy from the main pump). Each possible combination of net rail directions is considered and optimized. That is, for each possible net flow combination, the cost function in Eq. (9) is optimized separately using the appropriate main pump loss term from Eq. (4). Because the optimal solution will have one of these combinations of flow directions, we can take the solution obtained from using the net flow directions with the lowest cost to be the optimal solution of the whole problem.

In the case of zero net flow, a Lagrange multiplier is used to enforce this constraint. The constraint is that

$$\sum_{0}^{k_f} Q_i(k, d(k))\Delta t = 0 \tag{10}$$

The dual problem can be cast by augmenting the cost function to include a term $\lambda_i Q_i(k, d(k))$ for each rail i is to be constrained

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to zero net flow:

$$J_{DP}(d(\cdot)) = \sum_{k=0}^{k_f} \Bigg[Loss_{hyd,motor}(k, d(k)) + Loss_{e-motor}(k, d(k)) + \sum_{i=2}^{4} Loss_{MP,i}(k, d(\cdot)) + Loss_{switch}(k, d(k), , d(k+1)) + \sum_{\forall i \in I} \lambda_i Q_i(k, d(k)) \Bigg] \quad (11)$$

where I is the set of rails to be constrained to zero net flow. The value of each optimal λ_i is found by:

$$\lambda_i^* = \arg\max_{\lambda_i} [\min_{d(\cdot)} J(\lambda_i, d(\cdot))] \quad \forall i \in I \quad (12)$$

The inner loop is solved using dynamic programming [13]. The inner loop DP steps backwards in time with the decision variables as the CPR options and the propel motor displacements. Once the pressure rail and displacement choices have been made, the electric assist torque is mandated so that the drive cycle is met. For details on dynamic programming optimization for this problem, see [8]. As mentioned, this algorithm is not real-time implementable and is used only as a benchmark for the other real-time control laws.

4.2 Lagrange Multiplier Method

The Lagrange multiplier (LM) was originally studied because it provides a real-time implementable and computationally efficient way to calculate the optimal solution ignoring switching losses [7]. LM utilizes a simplified cost function (J_{LM}) which does not include switching losses:

$$J_{LM}(d(\cdot)) = \sum_{k=0}^{k_f} \Bigg[Loss_{hyd,motor}(k, d(k)) + Loss_{e-motor}(k, d(k)) + \sum_{i=2}^{4} Loss_{MP,i}(k, d(\cdot)) + \sum_{\forall i \in I} \lambda_i Q_i(k, d(k)) \Bigg] \quad (13)$$

With this simplification, the optimal solution can be found by performing a max-min optimization similar to (12) as long as the net flow directions are known. Compared to DP, the inner loop optimization can be solved at each time step as a static optimization.

For more details on the optimization framework, please see the original outline of the Lagrange multiplier method for this application [7].

This algorithm is real time implementable by first acquiring the net flow directions of each rail and the optimal Lagrange multiplier λ^* values by optimizing off-line using drive cycle data. Once λ^* is found for some drive cycle statistics, the inner loop optimization in (12) can be run in real time to determine the rail-selection. Thus, the architecture can be operated in similar drive cycles in real time, but estimates of losses in the pump/motors and motor/generators are required (along with the assumed rail flow directions and lambda values).

4.3 Greedy Control

DP is too computationally intensive and LM does not consider switching losses. Greedy control is investigated here as a computationally efficient way to include switching losses in the cost function. The greedy algorithm consider 1 step forward in time and evaluates as estimate of the instantaneous cost J_G given by:

$$J_G(k+1, d(k), d(k+1)) = Loss_{hyd,motor}(k+1, d(k+1)) + Loss_{e-motor}(k+1, d(k+1)) + \sum_{i=2}^{4} Loss_{MP,i}(k+1, d(k+1)) + Loss_{switch}(k+1, d(k), d(k+1)) + \sum_{\forall i \in I} \lambda_i Q_i(k+1, d(k+1)) \quad (14)$$

for each CPR and motor displacement option. Note that $d(k)$ is the previous chosen option and $d(k+1)$ is the next option to be chosen. The option resulting in the smallest cost at that time step is selected:

$$d^*(k+1) = arg \min_{d(k+1)} J_G(k+1, d(k), d(k+1)) \quad (15)$$

This algorithm always selects the action with the least current costly action which usually means it does not switch pressure rails unless it has to because the instantaneous switching losses often outweigh the efficiency losses in the other components. Unlike DP or LM, the greedy method does not consider the consequence of the current decision on future costs. This strategy has been previously proposed for use in a multi-pressure rail system for off highway vehicles[10].

Greedy control is real time implementable but requires models for the pump/motors, the motor/generators, and the losses due to switching.

4.4 Torque Minimizing Control

Torque minimizing control (TM) picks the CPR and displacement combination which result in hydraulic force/torques that are as close to the drive cycle demand as possible. This algorithm is not expected to perform as well as the others, but is useful for comparison and for investigating the effect of using smaller electric assist motors. The cost function J_{TM} is given as

$$J_{TM}(d(k+1)) = \|T_{elect}(d(k+1))\|_1 \quad (16)$$

where $T_{elect}(d(k+1))$ is a 1x4 vector containing the torques required by the electric modulation motor/generator for each actuator. The $\|\cdot\|_1$ denotes the one norm (the sum of absolute values of the elements). Since the decision for one actuator does not affect the cost function for the other actuator, this decouples the pressure rail decision of each actuator. This control also minimizes power through the electric modulation motors. Since energy losses themselves are not considered by this algorithm, frequent switches and the resulting switching losses are expected. Torque minimizing control is real time implementable and does not require any model for component efficiencies.

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4.5 Thresholding

A key component of the systems losses are switching losses, especially in the work circuit. An ad-hoc solution to reduce the number of switches is introduced here in which the current CPR selections are maintained, until the torque from the electric modulation motors passes some threshold, in which case a switch is made to the nearest force/torque rail. If the threshold is made to be zero, then the thresholding algorithm collapses to the same solution as the torque minimization control. The threshold here is set to 25 Nm because that is the maximum continuous torque that can be achieved by the motor generators at any speed. The motor/generators are capable of higher continuous torque but only at low speeds.

This control law is similar to the one for the Wavestar wave energy converter, which also defined a threshold between the desired force and the discrete hydraulic force exerted[12]. Thresholding is real time implementable and does not require any model for component efficiencies.

5. RESULTS

5.1 Illustrative Results

A sinusoidal drive cycle was first constructed to illustrate the types of decisions that each control law selects. In figure 3, this mock drive cycle is shown in light blue. The horizontal lines represent forces that can be achieved purely hydraulically. Purple dots represent the hydraulic force selected by each algorithm. The difference between the purple dot and the blue drive cycle curve is made up for by the small electric motor/generator. The modulation provided by this motor/generator is shown in figure 4. The same model electric motor/generator is used for lift, tilt and propel functions even though each function requires different amounts of torque. To provide fully continuous forces the lift and tilt functions each require at least 15 Nm from the electric motor/generators, while the propel tracks require 24 Nm. The model of electric motor used here is capable of providing 25 Nm of torque continuously at any speed and 50 Nm continuously at low speeds.

The Lagrange multiplier method does not take switching losses into account and so switches quite frequently, preferring to keep the modulation components operating at efficient operating conditions. Providing as much power as possible from the electric modulation motor/generator is preferred because of the increases in efficiency (ignoring losses due to switching). The dynamic programming algorithm limits the number of switches and utilizes the full capability of the electric modulation motor (50 Nm). The greedy algorithm also limits the number of switches but some switches are more costly than others, like ones taken with high actuator speeds and between far apart pressure rails. The torque minimizing control always selects the rail closest to the drive cycle torque. The electric torque is minimized, but there are frequent switching events. The thresholing control switches relatively often but this is because of the relatively low threshold given (25 Nm).

5.2 Drive Cycle Results

These algorithms were then enacted on actual drive cycle data from a 5 ton track loader executing a 22 minute basement dig drive cycle. The HHEA results were compared against an estimate of the fully electrified baseline system. Tables comparing the work and propel circuits of the HHEA using different CPR selection algorithms are shown in tables 1 and 2 respectively. The second to last row in each table shows the amount of energy required by the baseline system architecture. Regardless of control algorithm, the HHEA uses less energy, as seen by the positive energy savings percentages in the last row of the tables. The dynamic programming was the most efficient of all the algorithms because it is the global optimal solution.

For the work circuit, the biggest difference in the performance of each algorithm was the degree to which switching losses are limited. The number of switches, losses due to switching and ratio of positive work to input energy is shown in table 3 for each controller. DP, LM, and TM, refer to dynamic programming, Lagrange multiplier method, and torque minimizing control respectively. Generally, controllers with less switching events achieved higher ratios of positive output work to input energy (a measure of system efficiency). This is not always true as some switches cause more losses than others. The significant switching losses with the Lagrange multiplier method and the torque minimizing control are mostly due to the increased number of switching events. The greedy and thresholding algorithms had small numbers of switches and still achieved ratios of positive work to input work ratios of 0.96 and 0.81, respectively, over the electrified baseline work circuit.

In the propel circuit, the difference between control algorithms is shown in table 4. The differences were less pronounced than the work circuit and the relative performance of each algorithm was slightly different. Of the real time implementable control algorithms, the Lagrange multiplier method and the greedy algorithm performed well while the torque minimizing and thresholding algorithms performed the worst.

A comparison of results for both circuits are summarized in table 5. The ratio in the table compares the positive work done in the drive cycle to the net amount of energy provided by the battery. As the propel function accounts for most of the positive work done in the drive cycle, the propel circuit has a larger effect on the total ratio of positive work to input energy. The greedy algorithm performs close to the global optimal solution. The next best real-time controller was thresholding, which required no loss model.

Torque minimizing control was the worst control in both the work and propel circuits. Greedy control performed well in both circuits. Thresholding performed well in the work circuit but poorly in the propel circuit while the opposite is true for the Lagrange multiplier method. This is due to the high relative importance of throttling loss in the work circuit which the Lagrange multiplier method neglects but the thresholding method aims to minimize.

In the work circuit, the loss due to each switch was around 2.5 kJ while switches in the propel circuit are about 0.5 kJ. This is because there is much more fluid capacitance in the work circuit actuators than for propel. During a switch to pressures in the propel circuit, the volume in question is the hose volume which is very small compared with the cap side chamber of the work circuit actuators. This limits compressibility losses in the

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TABLE 1: ENERGY BREAKDOWN FOR THE WORK CIRCUIT OF THE HYBRID HYDRAULIC-ELECTRIC ARCHITECTURE (HHEA) USING DIFFERENT PRESSURE RAIL SELECTION ALGORITHMS FOR A BASEMENT DIG DRIVE CYCLE.

Work Circuit Energy [MJ]	Dynamic Programming	Lagrange Multiplier	Greedy	Torque Minimizing	Thresholding
Output Energy	3.41	3.41	3.41	3.41	3.41
Regen Energy	2.07	2.07	2.07	2.07	2.07
HECM Losses	0.89	1.00	1.29	0.46	0.68
Main Pump Losses	0.68	0.70	0.51	0.49	0.95
Switching losses	0.52	5.07	0.43	8.67	1.26
Input Energy	3.43	8.1	3.57	10.96	4.23
Baseline Energy Use	22.18	22.18	22.18	22.18	22.18
Energy Savings [%]	84.5	63.5	83.9	50.6	80.9

TABLE 2: ENERGY BREAKDOWN FOR THE PROPEL CIRCUIT OF THE HYBRID HYDRAULIC-ELECTRIC ARCHITECTURE (HHEA) USING DIFFERENT PRESSURE RAIL SELECTION ALGORITHMS FOR A BASEMENT DIG DRIVE CYCLE.

Propel Circuit Energy [MJ]	Dynamic Programming	Lagrange Multiplier	Greedy	Torque Minimizing	Thresholding
Output Energy	19.74	19.74	19.74	19.74	19.74
Regen Energy	2.91	2.91	2.91	2.91	2.91
HECM Losses	1.51	1.58	1.58	0.25	0.36
Main Pump Losses	3.95	3.55	3.79	6.28	6.37
Switching losses	2.44	2.83	2.69	3.30	3.08
Input Energy	29.30	29.32	29.48	32.99	32.95
Baseline Energy Use	34.74	34.74	34.74	34.74	34.74
Energy Savings [%]	15.7	15.6	15.1	5.0	5.2

TABLE 3: EFFECT OF SWITCHING LOSSES IN THE WORK CIRCUIT.

Work Circuit	DP	LM	Greedy	TM	Threshold
Number of Switches	652	3022	456	3049	701
Switch Loss [MJ]	0.5	5.1	0.4	8.7	1.3
Pos work / Input work	0.42	1.01	0.96	0.31	0.81

TABLE 4: EFFECT OF SWITCHING LOSSES IN THE PROPEL CIRCUIT.

Propel Circuit	DP	LM	Greedy	TM	Threshold
Number of Switches	4912	4536	4512	5366	4420
Switch Loss [MJ]	2.4	2.8	2.7	3.3	3.1
Pos work / Input work	0.67	0.67	0.67	0.60	0.60

propel circuit because there is less fluid being compressed. The pressure dynamics are also much faster, which further reduces the throttling loss in the propel circuit versus the work circuit.

Also, the same size of electric motor/generator was used for work and propel force/torque modulation but the torque requirement for propel is larger. The work circuit electric motor/generators are somewhat oversized, allowing for less frequent switching while pressure rails must be switched more often for the propel function, regardless of control algorithm.

Since switching losses play a smaller role in the propel circuit, the main difference between the algorithms is their minimization of the rest of the losses. The Lagrange and greedy control algorithms performed well in propel because they take the pump/motor and the motor/generator losses into account. Thresholding control only tries to limit switching events, which is an effective strategy only when the switching losses are dominant.

6. CONCLUSION

In this work, new real time control algorithms were presented and compared against previously studied pressure rail selection algorithms for the case of a compact track loader equipped with HHEA performinh a basement dig drive cycle. All the controllers gave energy saving in both the work and propel circuits compared with the electrified baseline architecture. The lowest energy savings was 23.4% and was given by utilizing the torque minimizing control on both work and propel. The highest (besides dynamic programming with 42.5%) was 41.9% energy savings and was given by utilizing the greedy control on both work and propel.

This work was completed only for the basement dig drive cycle. Further work is needed to test these algorithms on more drive cycles, but it is expected that the greedy algorithm will also perform near optimally. Also work is needed to validate results generated from this modeling effort using other simulation

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TABLE 5: RATIO BETWEEN POSITIVE WORK AND INPUT ENERGY FOR THE HYBRID HYDRAULIC-ELECTRIC ARCHITECTURE (HHEA) USING DIFFERENT PRESSURE RAIL SELECTION ALGORITHMS.

Pos work / Input work	DP	LM	Greedy	TM	Threshold
Work Circuit	1.01	0.42	0.96	0.31	0.81
Propel Circuit	0.67	0.67	0.67	0.60	0.60
Total Ratio	0.71	0.62	0.70	0.53	0.62

techniques and/or hardware in the loop testing.

Each controller requires different amount of information to operate. Dynamic programming requires full knowledge of the drive cycle ahead of time and is thus not suitable for real time operation. The Lagrange multiplier method requires models for losses in the pump/motors and motor/generators as well as information from off-line optimizations on similar drive cycles in order to get expressions for loss in the main pump. The greedy algorithm requires models for the pump/motors, the motor/generators, and the switching events. The torque minimizing and thresholding algorithms require no loss models.

The thresholding algorithm is a simple and effective real time operable control option if loss models are unavailable or are inaccurate. If trustworthy loss models are available, the greedy algorithm may give significant energy savings over thresholding control.

7. ACKNOWLEDGEMENT

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REFERENCES

[1] Lynch, Lauren and Zigler, Bradley T. "Estimating Energy Consumption of Mobile Fluid Power in the United States." Technical Report No. NREL/TP-5400-70240. National Renewable Energy Lab.(NREL), Golden, CO (United States). 2017.

[2] Volvo. "Electric Compact Excavator." Volvo (2023). Accessed 2023-07-31, URL https://www.volvoce.com/united-states/en-us/products/electric-machines/ecr25-electric/#overview.

[3] Bobcat. "All-Electric Compact Loaders." Bobcat (2023). Accessed 2023-07-31, URL https://www.bobcat.com/na/en/equipment/future-products/t7x-s7x-all-electric-compact-loaders.

[4] Caterpillar. "Growing a portfolio of electrified machines." Caterpillar (2023). Accessed 2023-07-31, URL https://www.cat.com/en_US/by-industry/mining/mining-responsibly/electrification.html.

[5] Li, Perry Y, Siefert, Jacob and Bigelow, David. "A hybrid hydraulic-electric architecture (HHEA) for high power off-road mobile machines." *Fluid Power Systems Technology*, Vol. 59339: p. V001T01A011. 2019. American Society of Mechanical Engineers.

[6] Li, Perry Y. "Hybrid Hydraulic-Electric Architecture (HHEA) for Mobile Machines (Final Report)." Technical Report No. DE-FOA-0001815. Department of Energy. 2022. DOI 10.2172/1878730. URL https://www.osti.gov/biblio/1878730.

[7] Siefert, Jacob and Li, Perry Y. "Optimal control of the energy-saving hybrid hydraulic-electric architecture (HHEA) for off-highway mobile machines." *IEEE Transactions on Control Systems Technology* Vol. 30 No. 5 (2021): pp. 2018–2029.

[8] Khandekar, Aditya, Wills, Jackson, Wang, Meng and Li, Perry Y. "Incorporating Valve Switching Losses into a Static Optimal Control Algorithm for the Hybrid Hydraulic-Electric Architecture (HHEA)." *Fluid Power Systems Technology*, Vol. 85239: p. V001T01A046. 2021. American Society of Mechanical Engineers.

[9] Vukovic, Milos, Sgro, Sebastian and Murrenhoff, Hubertus. "STEAM: A mobile hydraulic system with engine integration." *Fluid Power Systems Technology*, Vol. 56086: p. V001T01A005. 2013. American Society of Mechanical Engineers.

[10] Guo, Xiaofan, Lengacher, Jacob and Vacca, Andrea. "A Variable Pressure Multi-Pressure Rail System Design for Agricultural Applications." *Energies* Vol. 15 No. 17 (2022): p. 6173.

[11] Hansen, Rico H, Kramer, Morten M and Vidal, Enrique. "Discrete displacement hydraulic power take-off system for the wavestar wave energy converter." *Energies* Vol. 6 No. 8 (2013): pp. 4001–4044.

[12] Hansen, Rico H, Andersen, Torben O and Perdersen, Henrik C. "Analysis of discrete pressure level systems for wave energy converters." *Proceedings of the 2011 International Conference on Fluid Power and Mechatronics*: pp. 552–558. 2011. IEEE.

[13] Bellman, Richard. "Dynamic programming." *Science* Vol. 153 No. 3731 (1966): pp. 34–37.

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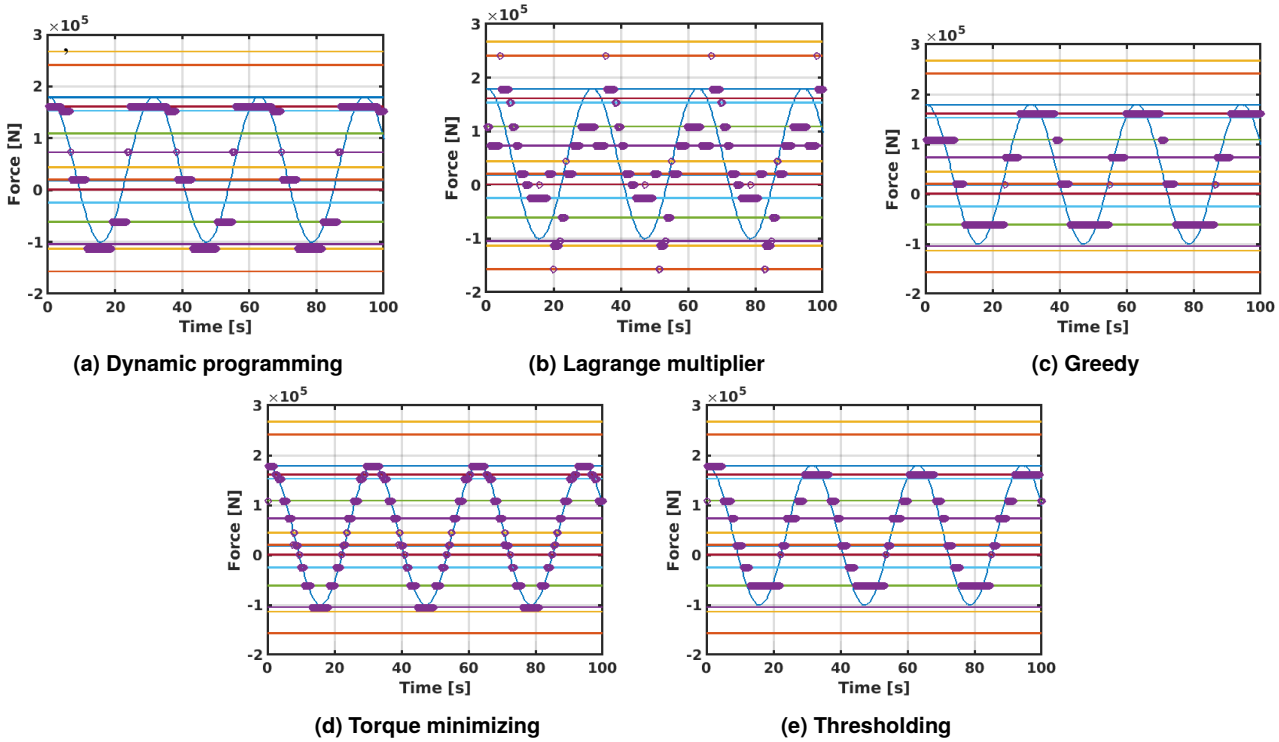

FIGURE 3: FORCE SELECTIONS USING DIFFERENT ALGORITHMS FOR AN IMAGINED DRIVE CYCLE.

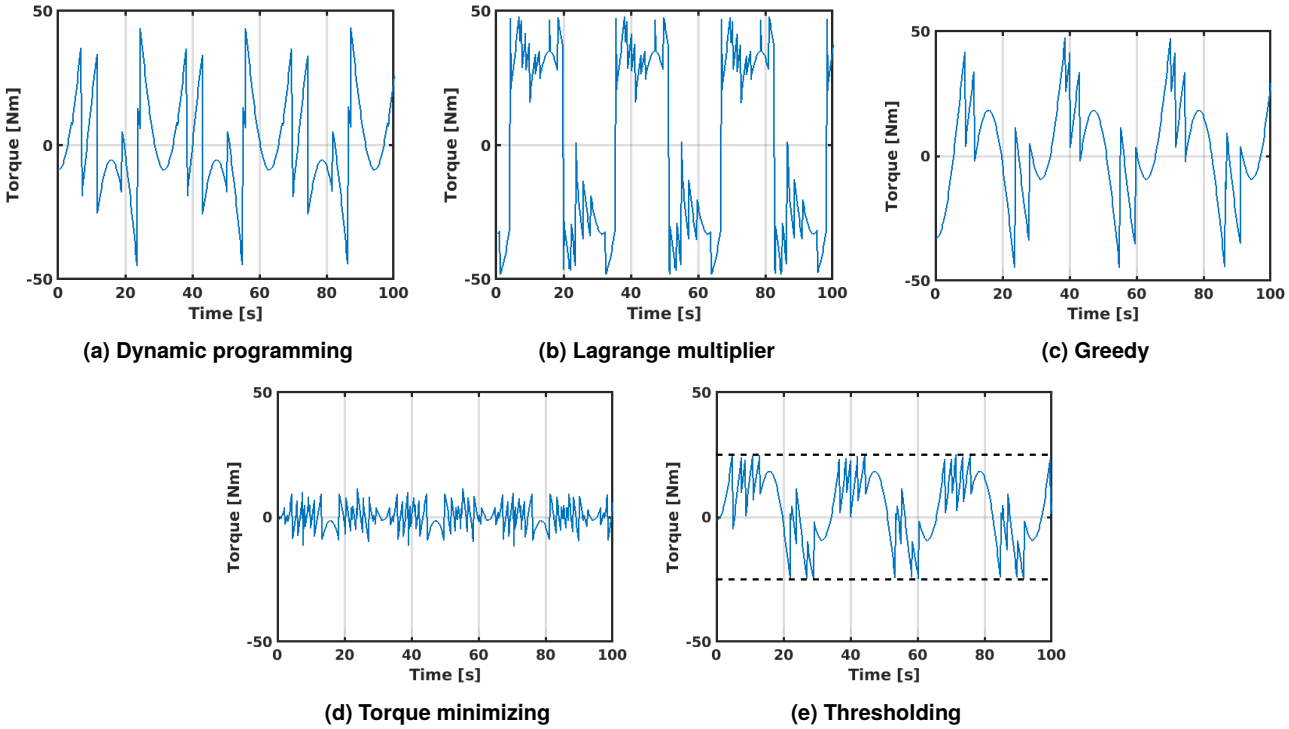

FIGURE 4: TORQUE REQUIRED BY ELECTRIC MODULATION MOTORS USING DIFFERENT PRESSURE RAIL SELECTIONS FOR AN IMAGINED DRIVE CYCLE

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